

Question:1

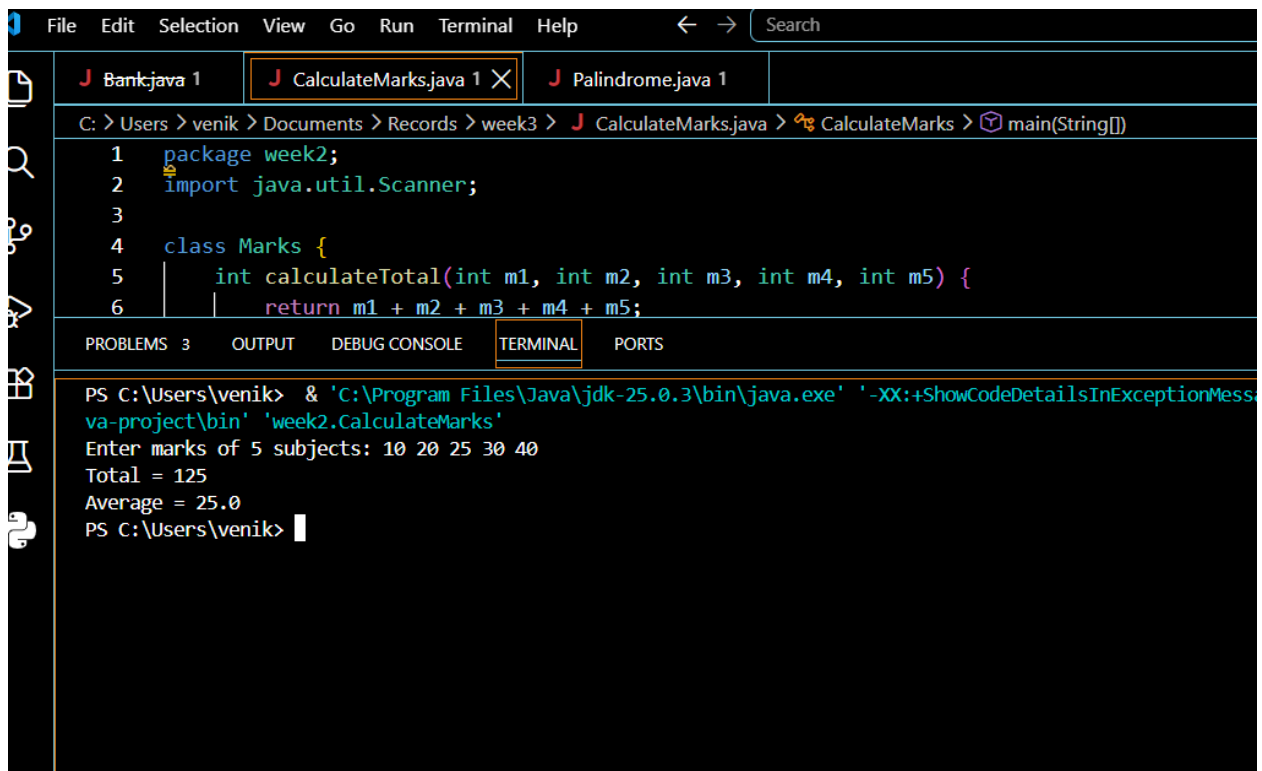
1. Write a java program which reads marks of five subjects through command line and print the total and average.

Steps:

1. Import the `Scanner` class.
2. Create a `Marks` class.
3. Define `calculateTotal()` to calculate the total marks of 5 subjects.
4. Define `calculateAverage()` to calculate the average marks.
5. Create the `CalculateMarks` class containing the `main()` method.
6. Create a `Scanner` object to accept input from the user.
7. Read marks of 5 subjects.
8. Create an object of the `Marks` class.
9. Call `calculateTotal()` and store the result in `total`.
10. Call `calculateAverage()` using the total and store the result in `average`.
11. Display the total marks.
12. Display the average marks.
13. Close the `Scanner`.
14. End the program.

Algorithm:

1. Start
2. Read the marks of five subjects: m_1 , m_2 , m_3 , m_4 , m_5 .
3. Create an object of the `Marks` class.
4. Calculate the total:
$$\text{Total} = m_1 + m_2 + m_3 + m_4 + m_5$$
5. Calculate the average:
$$\text{Average} = \text{Total} / 5$$
6. Display the total marks.
7. Display the average marks.
8. Close the input stream.
9. Stop.



The screenshot shows an IDE with three tabs: `Bank.java 1`, `CalculateMarks.java 1` (active), and `Palindrome.java 1`. The active tab displays the following Java code:

```
1 package week2;
2 import java.util.Scanner;
3
4 class Marks {
5     int calculateTotal(int m1, int m2, int m3, int m4, int m5) {
6         return m1 + m2 + m3 + m4 + m5;
7     }
8 }
9
10 public class CalculateMarks {
11     public static void main(String[] args) {
12         Scanner sc = new Scanner(System.in);
13         int m1 = sc.nextInt();
14         int m2 = sc.nextInt();
15         int m3 = sc.nextInt();
16         int m4 = sc.nextInt();
17         int m5 = sc.nextInt();
18         Marks m = new Marks();
19         int total = m.calculateTotal(m1, m2, m3, m4, m5);
20         double average = total / 5;
21         System.out.println("Total = " + total);
22         System.out.println("Average = " + average);
23     }
24 }
```

The bottom panel shows the terminal output:

```
PS C:\Users\venik> & 'C:\Program Files\Java\jdk-25.0.3\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMess
va-project\bin' 'week2.CalculateMarks'
Enter marks of 5 subjects: 10 20 25 30 40
Total = 125
Average = 25.0
PS C:\Users\venik> |
```

Question:2

1. Sample Java program to check whether the given number is prime palindrome or not.

Steps

1. Import the `Scanner` class.
2. Create a `Number` class.
3. Define a `check()` method to check whether a number is a palindrome.
4. Store the original number in a temporary variable.
5. Extract each digit using the modulus operator `%`.
6. Construct the reverse of the number.
7. Compare the original number with its reverse.
8. Return `1` if both are equal; otherwise, return `0`.
9. Create the `Palindrome` class with the `main()` method.
10. Get a number from the user.
11. Create an object of the `Number` class.
12. Call the `check()` method.
13. Display whether the number is a palindrome or not.
14. Close the `Scanner`.
15. End the program.

Algorithm

1. **Start**
2. Read the number `n`.
3. Store `n` in `temp`.
4. Initialize `reverse = 0`.
5. Repeat while `temp > 0`:
 - Find the last digit using `digit = temp % 10`.
 - Add the digit to the reverse using:
`reverse = reverse * 10 + digit`
 - Remove the last digit using:
`temp = temp / 10`
6. Compare `n` with `reverse`.
7. If `n == reverse`, return 1.
8. Otherwise, return 0.
9. If the returned value is 1, display **"number is a palindrome."**
10. Otherwise, display **"number is not a palindrome."**
11. Close the `Scanner`.
12. **Stop.**

FileEditSelectionViewGoRunTerminalHelp

Search

Bank.java 1

CalculateMarks.java 1

Palindrome.java 1

C: > Users > venik > Documents > Records > week3 > Palindrome.java > Number > check(int)

5class Number {

12reverse = reverse * 10 + digit;

13temp = temp / 10;

14}

15

16if (n == reverse) {

PROBLEMS3

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

PS C:\Users\venik> ^C

PS C:\Users\venik>

PS C:\Users\venik> & 'C:\Program Files\Java\jdk-25.0.3\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionStack' -va-project\bin' 'week2.Palindrome'

Enter a number: 12321

12321 is a palindrome.

PS C:\Users\venik>